Infrared Small Target Detection via Schatten Capped p Norm-Based Non-Convex Tensor Low-Rank Approximation

Fuju Yan, Guili Xu, Junpu Wang, Quan Wu, and Zhengsheng Wang

***Abstract*—For infrared (IR) small target detection, we propose a new spatial-temporal tensor (STT) decomposition model based on the tensor Schatten capped p norm (TSCpN) and total variation (TV) regularization. First, to explore spatial and temporal information, we construct an STT and introduce the prior weight map. Then, replacing the nuclear norm used to define tensor nuclear norm (TNN) with the nonconvex Schatten capped p (SCp) norm, we propose the TSCpN to approximate tensor rank and thus recover the low-rank background tensor. Next, to effectively eliminate background clutter from the sparse component, TV regularization is applied to constrain the sparse term. Finally, the proposed model is solved by the alternating direction method of multipliers (ADMM). Experimental results demonstrate the effectiveness and superiority of our proposed method in detecting IR small targets from various challenging sequences.**

***Index Terms*—Infrared small target detection, prior weight map, spatial-temporal tensor (STT), tensor Schatten capped p norm (TSCpN), total variation (TV) regularization.**

I. Introduction

THE study of infrared (IR) small target detection is significant for maritime surveillance systems and long-range stricks. However, since the targets are far away, their shape and texture information is often lacking. Moreover, targets are often obscured by complex backgrounds, for example, thick clouds, towering buildings, and trees. Hence, how to effectively extract small targets remains a crucial problem.

Recently, various approaches have been proposed for the detection of IR small targets. Particularly, the approach of tensor-based component analysis is a research hotspot. In practice, the detection performance of tensor component analysis-based methods mainly depends on: 1) how to make use of various useful information and 2) how to approximate the tensor rank.

On the one hand, to explore nonlocal self-correlationship, Dai and Wu [1] proposed the IR patch tensor (IPT) model. However, the patch-image model may destroy the structure of a small target. Moreover, the IPT model neglects temporal information that plays a vital role in capturing small targets in consecutive frames. Soon after, a spatial-temporal tensor (STT) [2], [3] model is proposed to excavate temporal information and evolved into the patch STT (PSTT) [4] model. Actually, the PSTT model is a combination of the IPT model and the STT model. So, the PSTT model may also face the risk of destroying the structure of a small target.

In addition, those reweighted methods also provide powerful tools to exploit useful information. Concretely, the reweighted methods take advantage of multiple prior weight maps combined with the sparse tensor to improve the detection performance. Therefore, the performance of those reweighted methods is inevitably affected by the quality of the prior weight map. Currently, [1], [5] employed two eigenvalues of the image structure tensor to calculate the prior weight map. However, the complex background is an obstacle to retaining the true target in the prior weight map. Without using an image structure tensor, Guan et al. [6] calculated the local contrast energy map as the prior weight map. Their experimental results show their method of obtaining the prior weight map is effective and efficient.

Hence, to mine useful information, we first construct an STT that allows us to exploit both spatial and temporal information. Then, to ensure the quality of the prior weight map and the efficiency of obtaining the prior weight map, the multiscale ring top-hat filter and area-based attribute filter are employed to calculate the prior weight map. In particular, to preserve the local details of a small target, all gray values of a region in a prior weight map are set to the same value.

On the other hand, to estimate the low-rank background tensor from an original tensor, tensor component analysis-based methods use various functions on singular values to approximate the tensor rank. Originally, the sum of the nuclear norm (SNN) is adopted [1]. However, the tensor low-rank approximation will be affected by the mode-1, 2, 3 unfolding matrices utilized to calculate the SNN. Different from the SNN, the tensor nuclear norm (TNN) is employed by Zhang et al. [7] to approximate the tensor rank and thus estimate the low-rank background tensor. But, due to the application of the nuclear norm which over-shrinks large singular values, TNN is treated as a poor function for estimating the low-rank background tensor. To ease it, Sun et al. [2], [3] replaced the nuclear norm used to define TNN with the weighted nuclear norm and

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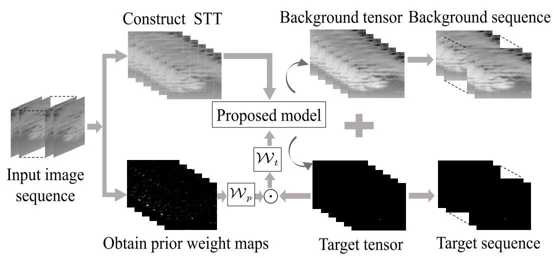

Fig. 1. Whole detection flowchart.

weighted Schatten p norm. Besides, other surrogate functions of tensor rank (partial sum of the TNN (PSTNN) [8], log operator-based tensor fibered nuclear norm (LogTFNN) [9], and the tensor capped nuclear norm (TCNN) [10], for example) are also developed to approximate the tensor rank and thus help improve detection performance. Here, inspired by the effectiveness and robustness of the nonconvex Schatten capped p (SCp) norm [11], we propose the tensor Schatten capped p norm (TSCpN) to approximate the tensor rank. Theoretically, TSCpN can not only penalize singular values smaller than a predefined threshold τ, but also shrink different singular values unequally with a parameter p, which brings the benefits of maneuverability and practicability in estimating the low-rank background tensor and helping improve detection performance. Particularly, to further improve the detection performance, we introduce total variation (TV) regularization into our tensor decomposition model to constrain the sparse term.

To sum up, we propose a new STT decomposition model based on the TSCpN and TV regularization (TSCpN-TV-STT) for IR small target detection. Our main contributions are listed as follows.

1) To make use of various useful information, the STT model is used to construct a tensor and the reweighted method is adopted to introduce the prior weight map computed from the multiscale ring top-hat filter, area-based attribute filter and an assimilation operation.
2) Different from existing surrogate functions used to approximate the tensor rank, the TSCpN is defined as a new surrogate function of tensor rank. By discussing the effects of main parameters of TSCpN, we demonstrate the maneuverability and practicability of the TSCpN in estimating the low-rank background tensor and helping improve detection performance. Also, TV regularization is employed to constrain the sparse term and thus help remove clutter from the detection results.
3) We solve the proposed model by using the alternating direction method of multipliers (ADMM). Experimental results validate the effectiveness and superiority of the proposed TSCpN-TV-STT model.

II. Methodology

Fig. 1 shows the procedure of detecting IR small targets. First, we construct an STT $\mathcal{X} \in R^{m\times n\times l}$ by sampling adjacent l images from an image sequence. Then, prior weight maps are calculated to define the weight tensor $\mathcal{W}_t$. Next, introducing the weight tensor $\mathcal{W}_t$ into the proposed tensor decomposition model, we extract the IR small target. Typically, the tensor decomposition model is built as follows:

$$\min_{\mathcal{L},\mathcal{T},\mathcal{N}} \operatorname{rank}(\mathcal{L}) + \lambda\|\mathcal{T}\|_1 + \beta f(\mathcal{T}) + \alpha\|\mathcal{N}\|_F^2 \quad \text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{T} + \mathcal{N} \tag{1}$$

where the rank($\mathcal{L}$) is used to identify the correlated backgrounds. $\|\mathcal{T}\|_1$ and $\|\mathcal{N}\|_F^2$ are employed to estimate sparse target tensor $\mathcal{T}$ and random noise $\mathcal{N}$, respectively. To realize constraint on the sparse term, the $f(\mathcal{T})$ is added. The parameters λ, β, and α are set to positive values to balance the above minimization model.

A. Proposed Model

1) Weight Tensor $\mathcal{W}_t$: To obtain the weight tensor $\mathcal{W}_t$, we calculate the prior weight map according to the following steps. First, we adopt the multiscale ring top-hat filter to capture varied-sized targets. Next, the area-based attribute filter is employed to reduce the influence of large-area clutter and single-pixel noise on capturing real targets. Finally, the assimilation operation is implemented, that is, all gray values of a region in the filtered image are set to the same value.

Once those prior weight maps are obtained, they will be combined with the absolute value of the iteratively updated sparse tensor $\mathcal{T}$ to define the weight tensor $\mathcal{W}_t$ ($\mathcal{W}_t = \mathcal{W}_p \odot 1/(|\mathcal{T}| + 0.01)$, the $\mathcal{W}_p$ denotes the reciprocal of those prior weight maps).

2) TSCpN-Based Regularization: Here, to approximate the tensor rank rank($\mathcal{L}$), we propose the TSCpN shown in (2) as a nonconvex surrogate of the tensor rank

$$\|\mathcal{L}\|_{\text{TSCpN}}^p = \frac{1}{n_3}\sum_{i=1}^{n_3}\sum_{j=1}^{r}\min\left(\sigma_j\left(\overline{\mathcal{L}}^{(i)}\right), \tau\right)^p. \tag{2}$$

Given a three-way tensor $\mathcal{L} \in R^{n_1\times n_2\times n_3}$, the r is the tensor tubal rank defined as $r = \operatorname{rank}_t(\mathcal{L})$ [2], [3]. The $\overline{\mathcal{L}}^{(i)}$ is the ith frontal slice of $\overline{\mathcal{L}}$ that is the discrete Fourier transformation (DFT) on $\mathcal{L}$ along the third dimension. The τ ($\tau > 0$) and p ($0 < p < 1$) are predefined parameters. It can be seen that TSCpN not only penalizes singular values smaller than the threshold τ, but also resorts to the parameter p to shrink different singular values unequally. Obviously, (2) shows the maneuverability and practicability of the TSCpN in protecting large singular values corresponding to the significant information of the low-rank background component, such as smooth zones, sharp edges, etc. Thus, TSCpN is able to perform more effectively in estimating smooth zones and strong edges to the low-rank background tensor and reducing false alarms in final detection results.

3) TV Regularization: Based on the differential operation, TV regularization is capable of capturing outliers of an image. Many efforts have applied TV regularization into a minimization model to facilitate image denoising. Thus, aiming to remove clutter from the sparse component, we introduce TV regularization into our minimization model

$$\|\mathcal{T}\|_{\text{TV}} = \|D(\mathcal{T})\|_1 = \|D_i\mathcal{T}\|_1 + \|D_j\mathcal{T}\|_1 \tag{3}$$

where D_i and D_j are differential operations along the horizontal direction i and the vertical direction j, respectively.

4) Proposed Model: By incorporating the $\mathcal{W}_t$, $\|\mathcal{L}\|_{\text{TSCpN}}^p$, and $\|\mathcal{T}\|_{\text{TV}}$ into the model (1), we obtain the proposed TSCpN-TV-STT model as follows:

$$\min_{\mathcal{L},\mathcal{T},\mathcal{N}} \|\mathcal{L}\|_{\text{TSCpN}}^p + \lambda\|\mathcal{W}_t \odot \mathcal{T}\|_1 + \beta\|\mathcal{T}\|_{\text{TV}} + \alpha\|\mathcal{N}\|_F^2 \quad \text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{T} + \mathcal{N}. \tag{4}$$

B. ADMM for the Proposed Model

First, by adding auxiliary variables $\mathcal{S}$ and $\mathcal{F}$, the model (4) is converted to the following formula:

$$\min_{\mathcal{L},\mathcal{T},\mathcal{N},\mathcal{S},\mathcal{F}} \|\mathcal{L}\|_{\text{TSCpN}}^p + \lambda\|\mathcal{W}_t \odot \mathcal{T}\|_1 + \beta\|\mathcal{F}\|_1 + \alpha\|\mathcal{N}\|_F^2$$
$$\text{s.t. } \mathcal{T} = \mathcal{S}, \quad \mathcal{F} = D(\mathcal{S}), \ \mathcal{X} = \mathcal{L} + \mathcal{S} + \mathcal{N}. \tag{5}$$

Then, we get the augmented Lagrangian function of (5)

$$\begin{aligned} L\big(&\mathcal{L}, \mathcal{T}, \mathcal{N}, \mathcal{S}, \mathcal{F}, \mathcal{M}_{1,2,3}, \mu\big) \\ &= \|\mathcal{L}\|_{\text{TSCpN}}^p + \lambda\|\mathcal{W}_t \odot \mathcal{T}\|_1 \\ &\quad + \beta\|\mathcal{F}\|_1 + \alpha\|\mathcal{N}\|_F^2 + \operatorname{tr}\left(\mathcal{M}_1^T(\mathcal{T} - \mathcal{S})\right) \\ &\quad + \operatorname{tr}\left(\mathcal{M}_2^T(\mathcal{F} - D(S))\right) \\ &\quad + \operatorname{tr}\left(\mathcal{M}_3^T(\mathcal{X} - \mathcal{L} - \mathcal{S} - \mathcal{N})\right) \\ &\quad + \frac{\mu}{2}(\|\mathcal{T} - \mathcal{S}\|_F^2 + \|\mathcal{F} - D(\mathcal{S})\|_F^2 \\ &\qquad + \|\mathcal{X} - \mathcal{L} - \mathcal{S} - \mathcal{N}\|_F^2) \end{aligned} \tag{6}$$

where $\mathcal{M}_{1,2,3}$ are the Lagrange multipliers, and μ is a positive penalty parameter. The problem (6) can be solved by iteratively and alternatively updating the following six subproblems.

1) $\mathcal{L}$ Subproblem:

$$\mathcal{L}^{k+1} = \arg\min_{\mathcal{L}} \frac{1}{\mu^k}\|\mathcal{L}\|_{\text{TSCpN}}^p + \frac{1}{2}\|\mathcal{L} - \mathcal{L}_{\text{temp}}\|_F^2 \tag{7}$$

where the $\mathcal{L}_{\text{temp}} = \mathcal{X} - \mathcal{S}^k - \mathcal{N}^k + (\mathcal{M}_3^k/\mu^k)$. The solution of (7) is shown in Algorithm 1. Especially, the scp operation [11] embedded into the tensor singular value thresholding (t-SVT) [12] framework can estimate a singular value σ_j with the relevant and known singular value δ_j

$$\operatorname{scp}(\delta_i, \tau, p) = \begin{cases} \sigma_j = \delta_j, & \tau < \tau_h \\ \sigma_j = \sigma_j^*, & \tau \geq \tau_h \end{cases} \tag{8}$$

where the $\tau_h = ((1/2c)(\sigma_j^* - \delta_j)^2 + (\sigma_j^*)^p)^{1/p}$, the $\sigma_j{}^* = GST(\delta_j, c, p)$. The generalized soft-thresholding algorithm (GST) is the GST presented in [13].

Algorithm 1 Solution of (7)

Input: $\mathcal{L}_{temp} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, *parameter* τ, p
Output: $\mathcal{L}$
1. Conduct FFT operation: $\bar{\mathcal{L}}_{temp} = fft\big(\mathcal{L}_{temp}, [], 3\big)$.
2. Compute each frontal slice of $\bar{D}$ by:
for $i = 1, \ldots, \left[\frac{n_3+1}{2}\right]$ **do**
  $[\mathcal{U}, \mathcal{S}, \mathcal{V}] = \operatorname{SVD}\left(\bar{\mathcal{L}}_{temp}^{(i)}\right)$, $S = \operatorname{diag}(\delta_1, \ldots, \delta_r)$;
  for $i = 1, \ldots, r$ **do**
    $\sigma_i = scp(\delta_i, \tau, p)$;
  end
  $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$
  $\bar{D}^{(i)} = U \cdot \Sigma \cdot V^*$
end
for $i = \left[\frac{n_3+1}{2}\right], \ldots, n_3$ **do**
  $\bar{D}^{(i)} = conj\left(\bar{D}^{(n3-i+2)}\right)$
end
3. Compute $\mathcal{L} = ifft\big(\bar{D}, [], 3\big)$

2) $\mathcal{T}$ Subproblem:

$$\mathcal{T}^{k+1} = \arg\min_{\mathcal{T}} \frac{\lambda}{\mu^k}\|\mathcal{W}_t^k \odot \mathcal{T}\|_1 + \frac{1}{2}\|\mathcal{T} - \mathcal{T}_{\text{temp}}\|_F^2 \tag{9}$$

where $\mathcal{T}_{\text{temp}} = \mathcal{S}^k - (\mathcal{M}_1^k/\mu^k)$. The $\mathcal{T}^{k+1}$ can be solved by the tensor soft-threshold shrinkage operator, $\mathcal{T}^{k+1} = \operatorname{soft}(\mathcal{T}_{\text{temp}}, \ \lambda\mathcal{W}_t^k/\mu^k)$, $\mathcal{W}_t^k = \mathcal{W}_p \odot (1/(|\mathcal{T}^k| + 0.01))$.

3) $\mathcal{S}$ Subproblem:

$$\mathcal{S}^{k+1} = \arg\min_{\mathcal{S}} \frac{\mu^k}{2}\big(\|\mathcal{T}^{k+1} - \mathcal{T}_{\text{temp}}\|_F^2 + \|\mathcal{F}^k - \mathcal{F}_{\text{temp}}\|_F^2 + \|\mathcal{S}_{\text{temp}} - S\|_F^2\big) \tag{10}$$

where $\mathcal{F}_{\text{temp}} = D(\mathcal{S}) - (\mathcal{M}_2^k/\mu^k)$, $\mathcal{S}_{\text{temp}} = \mathcal{X} - \mathcal{L}^{k+1} - \mathcal{N}^k + (\mathcal{M}_3^k/\mu^k)$. To solve $\mathcal{S}^{k+1}$, (10) can be solved by using the fast Fourier transform (FFT). The solution is expressed as

$$\mathcal{S}^{k+1} = \operatorname{ifftn}\frac{\operatorname{fftn}(\text{temp})}{2\mu^k\mathcal{I} + \mu^k(\operatorname{fftn}(D))^2} \tag{11}$$

where $\text{temp} = \mu^k\mathcal{T}^{k+1} + \mathcal{M}_1^k + D^T(\mu^k\mathcal{F}^k + \mathcal{M}_2^k) + \mu^k(\mathcal{X} - \mathcal{L}^{k+1} - \mathcal{N}^k) + \mathcal{M}_3^k$. The $\mathcal{I}$ is the unit tensor. The fftn and ifftn denote the FFT and its inverse transform. D^T represents the adjoint operator of D.

4) $\mathcal{F}$ Subproblem:

$$\mathcal{F}^{k+1} = \arg\min_{\mathcal{F}} \frac{\beta}{\mu^k}\|\mathcal{F}\|_1 + \frac{1}{2}\|\mathcal{F} - \mathcal{F}_{\text{temp}}\|_F^2. \tag{12}$$

Similar to the $\mathcal{T}$ subproblem, the $\mathcal{F}$ subproblem is also solved by the soft-threshold shrinkage operator, $\mathcal{F}^{k+1} = \operatorname{soft}(\mathcal{F}_{\text{temp}}, \beta/\mu^k)$.

5) $\mathcal{N}$ Subproblem:

$$\mathcal{N}^{k+1} = \arg\min_{\mathcal{N}} \frac{\alpha}{\mu^k}\|\mathcal{N}\|_F^2 + \frac{1}{2}\|\mathcal{N} - \mathcal{N}_{\text{temp}}\|_F^2 \tag{13}$$

where the $\mathcal{N}_{\text{temp}} = \mathcal{X} - \mathcal{L}^{k+1} - \mathcal{S}^{k+1} + (\mathcal{M}_3^k/\mu^k)$. The solution of (13) is expressed as:

$$\mathcal{N}^{k+1} = (2\alpha + \mu^k)^{-1}(\mu^k\mathcal{N}_{\text{temp}}). \tag{14}$$

6) Multiplier Subproblem:

$$\begin{aligned} \mathcal{M}_1^{k+1} &= \mathcal{M}_1^k + \mu^k\big(\mathcal{T}^{k+1} - \mathcal{S}^{k+1}\big) \\ \mathcal{M}_2^{k+1} &= \mathcal{M}_2^k + \mu^k(\mathcal{F}^{k+1} - D(\mathcal{S}^{k+1})) \\ \mathcal{M}_3^{k+1} &= \mathcal{M}_3^k + \mu^k\big(\mathcal{X} - \mathcal{L}^{k+1} - \mathcal{S}^{k+1} - \mathcal{N}^{k+1}\big) \\ \mu^{k+1} &= \min\{\rho\mu^k, \mu_{\max}\}. \end{aligned} \tag{15}$$

Summarizing the above six steps, we show the whole optimization procedure in Algorithm 2.

III. Experimental Results and Analyses

Experiments on six IR sequences are adopted to test the effectiveness of the proposed TSCpN-TV-STT model. Furthermore, the six different methods including top-hat filter [14], multiscale local contrast measure using local energy factor (MLCM-LEF) [15], non-negative infrared patch-image model based on partial sum minimization of singular values (NIPPS) [16], reweighted infrared patch-tensor model (RIPT) [1], PSTNN [8], and edge and corner awareness-based spatial–temporal tensor model (ECA-STT) [5] are chosen for comparison. In practice, the parameter settings are manually adjusted according to the actual detection results. Referring to the work in [3], [5], we use the signal-to-clutter ratio gain

Algorithm 2 ADMM for Solving the Proposed Model (4)

Input: IR image sequence, parameters $\lambda, \beta, \alpha, \tau, p$;
Initial $\mathcal{L}^0, \mathcal{T}^0, \mathcal{S}^0, \mathcal{F}^0, \mathcal{N}^0, and\ \mathcal{M}^0_{1,2,3}$ are initialized to 0, $\mu^0 = 10^{-3}$, $\rho = 1.5$, $\varepsilon = 10^{-7}$;
while *not converged* **do**
  Update $\mathcal{L}^{k+1}$ in (7) with the Algorithm. 1;
  Update $\mathcal{T}^{k+1}$ in (9) using the soft thresholding operator;
  Update $\mathcal{S}^{k+1}$ in (10) using the FFT in (11);
  Update $\mathcal{F}^{k+1}$ by (12);
  Compute $\mathcal{N}^{k+1}$ in (13) via (14);
  Compute Lagrange multiplier $\mathcal{M}^{k+1}_{1,2,3}$ and penalty parameter μ^{k+1} by (15);
  $k = k + 1$;
  Verify the stopping criteria: $\frac{\|\mathcal{X}-\mathcal{L}^k-\mathcal{S}^k-\mathcal{N}^k\|_F}{\|\mathcal{X}\|_F} < \varepsilon$
end
Output: $\mathcal{L}, \mathcal{T}, \mathcal{N}$

(SCRG), background suppression factor (BSF), and contrast gain (CG) to quantify the detection results. Notably, unlike the method used to measure CG in the papers [3], [5], we express CG as

$$\mathrm{CG} = \frac{C_{\det}}{|C_{\mathrm{in}}| + \epsilon} \tag{16}$$

where the C is defined as: $C = (M_t - M_b)/(M_t + M_b)$. The M_t and M_b are the maximum values of an internal window and a surrounding window. The internal window and surrounding window are included in a local window of size 60×60. Besides, the receiver operating characteristic (ROC) curve and the area under the curve (AUC) [17] are also shown. In practice, larger SCRG, BSF, CG, and AUC suggest that the proposed method produces more satisfactory detection results.

A. Parameter Analysis and Parameter Settings

The effects of parameters λ, β, α in the minimization model (4) on detection performance are analyzable. To realize the minimization of the model (4), larger λ and β will produce smaller $\mathcal{T}$, which suggests we could suppress background clutter by setting λ and β to larger values. Similarly, the larger α could reduce the composition of the component $\mathcal{N}$, which in turn increases the composition of other components, thereby leaving more background clutter in the sparse component. However, since the component $\mathcal{N}$ contains both positive and negative values, the effect of parameter α may not be obvious.

Next, we mainly discuss the effects of parameters τ and p on the detection performance. Note that in order to make the following parameter analysis more convincing and concise, we prepare a new sequence named Seq. 1'–6' that includes six new sequences in advance. Briefly, the six new sequences are constructed by sampling one image every three frames from the above-mentioned six sequences. From Fig. 2(a), it can be observed that the increased τ will decrease the AUC. Thus, a smaller τ will perform better in removing clutter from the sparse component. However, it can be known from (2) and the first branch of (8) that a too small τ can preserve most of the singular values, which may result in the small target will be left in the low-rank component. Therefore, although a smaller parameter τ is beneficial to remove clutter from the sparse component, the value of parameter τ should not be set too small. Here, we set parameter τ to 200. Clearly, when τ is set to 200, the AUC retains a larger value and does not drop much compared to τ equal to 50 and 100. As for parameter p, from Fig. 2(b), it can be seen that the increased parameter p will reduce the AUC, which means a smaller p will perform better in suppressing background clutter. Here, to prevent too small p from wiping out real targets in detection results, we set parameter p to 0.85. Clearly, when p is set to 0.85, the AUC retains a larger value and does not drop much compared to p equal to 0.75 and 0.8. Particularly, referring to [5], we define the l as 3.

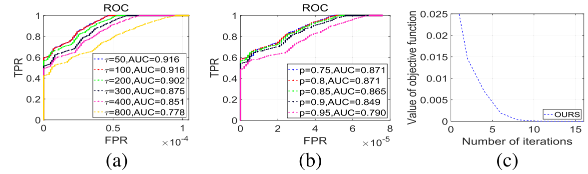

Fig. 2. ROC curves of different parameters and the convergence curve of the proposed algorithm. (a) Different τ values. (b) Different p values. (c) Convergence curve.

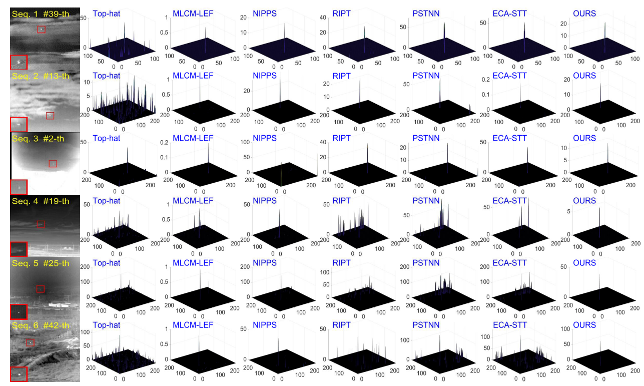

Fig. 3. Representative detection results of all comparison algorithms on Seq. 1–6.

B. Convergence Analysis

By referring to the work in papers [3], [5], Fig. 2(c) shows the empirical convergence analysis of the proposed method. It can be seen that the objective function drops rapidly to close to 0, which means the proposed method has good convergence.

C. Experimental Comparisons

1) Vision Comparison: We show detection results of representative frames of all sequences in Fig. 3. Obviously, the top-hat filter can not effectively remove background clutter from the detection results while preserving the real target. Unlike the top-hat filter, the MLCM-LEF performs well in suppressing clutter and highlighting targets. However, by observing the detection results of Seq. 4 and Seq. 5, we realize that MLCM-LEF is limited in enhancing real targets disturbed by similar background clutter. Besides, we can see the wide black edge on the representative frame of Seq. 3 interferes with the detection performance of the NIPPS. Besides, when the RIPT, PSTNN, and ECA-STT are used to separate targets from the nonuniform background, they will leave clutter in the detection results. By contrast, the last column of Fig. 3 shows the proposed method is superior to other methods in suppressing background clutter and separating true targets from various challenging sequences.

2) Quantitative Comparison: Comparing the quantitative results of the last column with the results of other columns, we draw a conclusion that our method is competitive among all comparison algorithms. It is clear that in the last column of Table I, there are more bolded values representing the best

TABLE I

QUANTITATIVE RESULTS OF ALL COMPARISON ALGORITHMS ON SEQ. 1–6

		Top-hat	MLCM-LEF	NIPPS	RIPT	PSTNN	ECA-STT	OURS
Seq. 1 [18] (52 frames)	$\overline{\mathrm{SCRG}}$	208.68	276.51	865.27	499.57	914.99	231.02	**937.37**
	$\overline{\mathrm{BSF}}$	20.86	173.19	166.82	90.54	166.44	128.37	**176.59**
	$\overline{\mathrm{CG}}$	1.43	3.35	3.13	2.46	3.26	3.15	**3.36**
Seq. 2 (70 frames)	$\overline{\mathrm{SCRG}}$	332.80	621.90	1065.66	**1570.79**	1514.48	924.54	1254.47
	$\overline{\mathrm{BSF}}$	38.85	81.89	85.29	89.96	90.97	89.97	91.08
	$\overline{\mathrm{CG}}$	2.31	2.83	2.86	2.83	2.88	2.82	2.87
Seq. 3 (90 frames)	$\overline{\mathrm{SCRG}}$	5736.07	4736.35	1735.68	**10222.46**	4828.99	4190.99	10078.64
	$\overline{\mathrm{BSF}}$	157.76	211.30	56.76	151.14	93.92	95.59	**213.77**
	$\overline{\mathrm{CG}}$	3.01	**3.18**	1.92	2.58	2.12	2.35	**3.18**
Seq. 4 (150 frames)	$\overline{\mathrm{SCRG}}$	3424.72	5338.84	3291.10	**15795.52**	12771.37	6703.68	12412.03
	$\overline{\mathrm{BSF}}$	36.59	77.50	50.22	**77.93**	**77.93**	**77.93**	**77.93**
	$\overline{\mathrm{CG}}$	2.11	**2.34**	2.07	**2.34**	**2.34**	**2.34**	**2.34**
Seq. 5 [19] (150 frames)	$\overline{\mathrm{SCRG}}$	1991.90	8892.35	5525.57	15626.68	16230.28	12464.50	**17291.31**
	$\overline{\mathrm{BSF}}$	6.85	37.17	16.40	27.29	**37.95**	37.84	37.71
	$\overline{\mathrm{CG}}$	1.14	1.54	1.35	1.39	**1.55**	**1.55**	1.54
Seq. 6 (150 frames)	$\overline{\mathrm{SCRG}}$	187.93	576.34	1240.96	439.85	1460.45	289.98	**1612.66**
	$\overline{\mathrm{BSF}}$	16.15	85.04	64.92	30.90	57.52	38.07	**104.96**
	$\overline{\mathrm{CG}}$	1.24	2.57	1.61	0.86	1.73	1.34	**2.67**

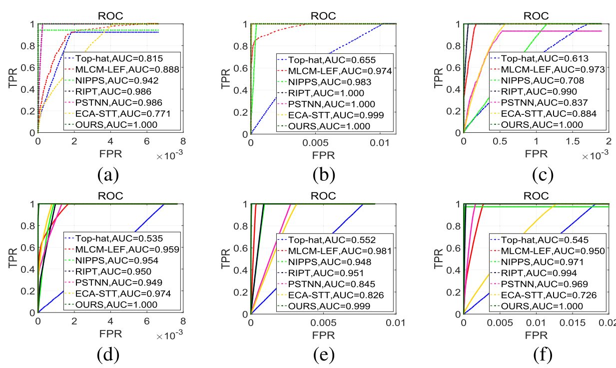

Fig. 4. ROC curves of all comparison algorithms on Seq. 1–6.

TABLE II

QUANTITATIVE RESULTS OF DIFFERENT MODELS ON SEQ. 1'–6'

	Models	$\overline{\mathrm{SCRG}}$	$\overline{\mathrm{BSF}}$	$\overline{\mathrm{CG}}$
TNN-STT [7]	$\min\limits_{\mathcal{L},\mathcal{T},\mathcal{N}} \|\mathcal{L}\|_{TNN} + \lambda\|\mathcal{W}_t \odot \mathcal{T}\|_1 + \alpha\|\mathcal{N}\|_F^2$	6832.20	64.83	1.92
TSpN-STT [3]	$\min\limits_{\mathcal{L},\mathcal{T},\mathcal{N}} \|\mathcal{L}\|_{TSpN}^p + \lambda\|\mathcal{W}_t \odot \mathcal{T}\|_1 + \alpha\|\mathcal{N}\|_F^2$	7914.86	74.68	2.08
TSCpN-STT	$\min\limits_{\mathcal{L},\mathcal{T},\mathcal{N}} \|\mathcal{L}\|_{TSCpN}^p + \lambda\|\mathcal{W}_t \odot \mathcal{T}\|_1 + \alpha\|\mathcal{N}\|_F^2$	8242.87	79.52	2.14
OURS	$\min\limits_{\mathcal{L},\mathcal{T},\mathcal{N}} \|\mathcal{L}\|_{TSCpN}^p + \lambda\|\mathcal{W}_t \odot \mathcal{T}\|_1 + \beta\|\mathcal{T}\|_{TV} + \alpha\|\mathcal{N}\|_F^2$	8593.26	100.67	2.48

evaluation result in each row. Specifically, there are many identical values in the same row, which can be attributed to the detection results normalized to 0–255. Furthermore, from Fig. 4, it can be seen that our method yields the largest AUC, which further demonstrates its superiority and effectiveness.

3) Ablation Experiment: In order to demonstrate the effect of important components, we compare our model with TNN-STT [7] model, TSpN-STT [3] model, and TSCpN-STT model on the Seq. 1'–6'. From Table II, it is obvious that the TSCpN-STT model compared to the TNN-STT model and the TSpN-STT model brings better quantitative results, which indicates the validity of the TSCpN. Also, by comparing the quantitative results of TSCpN-STT model with those of our proposed model, we realize that the $\overline{\mathrm{SCRG}}$, $\overline{\mathrm{BSF}}$, and $\overline{\mathrm{CG}}$ of our proposed method increase by 4.25%, 26.60%, and 15.89%, respectively. The increased $\overline{\mathrm{SCRG}}$, $\overline{\mathrm{BSF}}$ and $\overline{\mathrm{CG}}$ indicate that TV regularization plays an important role in improving detection performance.

IV. CONCLUSION

In this letter, we proposed the TSCpN-TV-STT model for IR small target detection. Different from other surrogate functions of tensor rank, TSCpN was proposed to recover the low-rank background tensor. By adjusting the parameters τ and p, we have shown the maneuverability and practicability of TSCpN in estimating the low-rank background tensor and reducing the false alarm in detection results. The STT and the prior weight map were applied to excavate useful information for capturing true targets. TV regularization, which helps suppress clutter in detection results, was employed to constrain the sparse term. Extensive experimental results have demonstrated our proposed method is competitive with other state-of-the-art methods.

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